PHYSICS 535/2&3
COLLECTION OF PRACTICAL QUESTIONS
O-LEVEL
Practical items

ITEM ONE (1):

In one of the schools, a learner in class finds difficulty while reading the words written on the chalkboard. The optician discovered long sight in the learner and advised it can be corrected by wearing spectacles with convex lenses of power between $+\mathbf{8}.\mathbf{3}\boldsymbol{D}$ and $+\mathbf{12}.\mathbf{5}\boldsymbol{D}$ inclusive. The parent of the learner bought spectacles whose lenses had no power indicated on them and approached you for assistance.

TASK:

You are provided with a convex lens from the laboratory with identical properties as the one bought by the parent, advise on its suitability in the spectacles.

HINT: In your report include the following;

Aim, variables (independent, dependent and controlled), hypothesis, procedure, errors anticipated, precautions to mitigate the errors, present data accurately (at least 5 sets), analyse data (using graph) and advice/recommend.

ITEM TWO (2):

Some of the community members complains about the new dry cells, called Pana-energy sold in the shops taking only a short time and get used up. One members said that the pana-energy dry cells have higher internal resistance and that a standard dry cell has an internal resistance in the range of $\mathbf{0}.\mathbf{2}\Omega$ to $\mathbf{0}.\mathbf{8}\Omega$. Another member brings to you a new dry cell of the same type bought from the shop to discover the truth.

TASK:

You are provided with one dry cell from the laboratory with similar properties like the one bought from the shop by the second member, comment on its quality.

HINT: In your report include the following;

Aim, variables (independent, dependent and controlled), hypothesis, procedure, errors anticipated, precautions to mitigate the errors, present data accurately (at least 5 sets), analyse data (using graph) and advice/recommend.

ITEM THREE (3):

One afternoon, the school secretary was drilling holes on the papers for filing using a heavy-duty punching machine, but suddenly the punching machine stopped working. The mechanic discovered that the spring of force constant between $\mathbf{200}\boldsymbol{Nm}-\mathbf{1}$ and $\mathbf{600}\boldsymbol{Nm}-\mathbf{1}$, in the punching machine was damaged and required replacement. The secretary bought a new spring from a nearby hardware, but without specification and contacted you for help.

TASK:

You are provided with a spring from the laboratory, with identical properties as the one bought by the secretary and whose force constant is ***twenty times*** smaller than that in the punching machine, advice the secretary on its suitability for use in the punching machine.

HINT: Include the following in your report;

Aim, variables (independent, dependent and controlled), hypothesis, procedure, errors anticipated, precautions to mitigate the errors, present data accurately (at least 5 sets), analyse data (using graph) and advice/recommend.

ITEM FOUR (4):

A builder fixing windows for a community member proposed that the best glass for the window shutter should be of critical angle in the range of **38°** to **42°** inclusive. The community member moved to a nearby shop that sells glasses and bought a large sheet of glass but was not sure of its critical angle and sought for assistance from you.

TASK:

You are provided with a glass block from the laboratory, with similar properties as the large sheet of glass bought by the community member, advise accordingly on its fitness for window shutter.

HINT: Include the following in your report;

Aim, variables (independent, dependent and controlled), hypothesis, procedure, errors anticipated, precautions to mitigate the errors, present data accurately (at least 5 sets), Analyse data (using graph) and advice/recommend.

ITEM FIVE (5)

In a rural community where electricity supply is inconsistent, residents depend heavily on a battery-powered radio for accessing news broadcasts, weather updates, educational programmes, and government announcements. Recently, the community radio has been malfunctioning—switching off unexpectedly even when fitted with newly purchased dry cells from the local village shop. This has caused significant inconvenience and raised suspicions about the quality of dry cells being sold locally. One of the concerned community members, travelled to a well-known electronics shop in the nearby town and purchased a pair of dry cells, hoping they would be more reliable. However, before using them in the community radio, he wishes to verify whether these cells are indeed of good quality and will last for a reasonable duration. A good-quality dry cell is expected to have an electromotive force (e.m.f.) within the range of 1.2 V to 1.8 V and a low internal resistance to ensure sustained performance.

TASK:

You are provided with a dry cell similar to the community member purchased verify whether the cell meets the standard for a good-quality dry cell.

ITEM SIX (6)

A man was driving his car through a severe traffic jam on Kampala Road when a motorcyclist accidentally collided with the front of his vehicle, damaging one of the car's headlamps. The internal concave mirror within the headlamp assembly was also broken and needed replacement. According to the car's log book, the original concave mirror had a focal length within the range of 8 cm to 23 cm. The man took his car to a nearby garage for repairs. The mechanic there has a spare concave mirror, but its focal length is unknown, and therefore he is unsure whether it is suitable for use in the car's headlamp assembly. The mechanic has approached you for assistance in identifying the exact focal length of the mirror.

TASK:

You are provided with the concave mirror available at the garage. As a physics learner, help the mechanic.

ITEM SEVEN (7)

A technician was repairing a faulty radio set and needed a resistor with a resistance of 2Ω. He picked one resistor from a newly purchased pack that was marked 2Ω, but after installing it, the radio still failed to work properly. The technician suspects that the resistor might have been wrongly labelled during packaging at the factory. This time he has acquired another resistor from a different source. Before using it, he wants to be sure that it has a correct resistance needed to repair the radio. However, the technician lacks the device for measuring the actual resistance of the resistor and has contacted you for assistance.

TASK:

You are provided with a resistor having similar property as the one the technician acquired, confirm whether it is suitable for use in the radio.

ITEM EIGHT (8)

An entertainment hall owner uses a projector to display images on a screen. During the hall's setup, the projector lens accidentally fell and broke. The projector manual states that the appropriate lens should have a focal length between 8.0cm and 12.0 cm. A replacement lens has been provided, but its suitability is uncertain.

TASK:

You have been asked to apply your knowledge of physics to investigate and determine whether the replacement lens meets the required specifications and can be used in the projector.

ITEM NINE (9)

Recently, more students have been receiving inaccurate results in their Physics experiments. They suspect that the internal resistance of the cells being used might be higher than the acceptable range of 0.8Ω to 1.5Ω. However, the students are unsure how to verify this. They have asked for your assistance.

TASK:

Use your understanding of Physics to investigate and determine whether the internal resistance of the cells falls within the acceptable range and if the cells are suitable for use in practical work.

ITEM TEN (10)

During a bicycle riding competition, the shock absorber of the Sports bike of your friend got damaged and the mechanic advised him to purchase a spring of force constant ranging from 60,000Nm-1 to 120,000Nm-1 to replace it. Your friend bought a spring of unknown force constant, so he needed to confirm if it is fit for replacement in the damaged shock absorber but didn't have the sufficient knowledge to do so, and approached you for help. Hint. The spring constant of the spring required to be replaced in the shock absorber is 3000 times that of the spring available in the physics laboratory.

TASK:

You are provided with a spring from the physics laboratory; use it to advise your friend.

ITEM ELEVEN (11)

During a physics lesson, the teacher used a projector to display the information he was explaining to students, unfortunately one of the students accidentally destabilized the table where the projector was placed and it fell down. When the teacher was trying to put back the projector in

position, he discovered that its' lens of power ranging from 8.0D to 12.0D got damaged and needed to be replaced. The teacher tasked the student who caused the damage to immediately go and buy the lens to be replaced in the projector. The student went to the shop and bought a lens but was not sure whether it was fit for replacement in the damaged projector.

TASK:

You are provided with a lens with similar properties as the one bought by the student, use it to advise the student.

ITEM TWELVE (12)

Your parent is constructing a new house in the village and he was advised by the Engineer to buy a glass of refractive index ranging from 1.4 to 1.7 to fit in the windows. He went and bought glasses but was not sure if it was of the required refractive index, so he approached you for help.

TASK:

You are provided with a glass block with similar properties as the glass material bought by your parent, use it to advise your parent accordingly.

ITEM THIRTEEN (13)

During a Physics practical lesson, Mukisa, one of the students in the class broke a metre rule of mass ranging from 70g to 150g. As the school policy was, he was told to buy a similar metre rule to replace the broken one. Mukisa went and bought a metre rule but was not sure if it was fit for replacement of the damaged metre rule so he approached you for help.

TASK:

You are provided with a metre rule with similar properties as the one bought by Mukisa, use it to advise him accordingly.

ITEM FORTEEN (14)

Your uncle's flat iron got damaged and he was advised by the mechanic to purchase a resistance wire of resistance ranging from 10 to 65 to replace the damaged one. Your uncle purchased a resistance wire of unknown resistance and didn't have the sufficient knowledge to determine it, so he approached you for help.

TASK:

You are provided with a resistance wire of similar properties as the one bought by your uncle, use it to advise him accordingly.

ITEM FIFTEEN (15)

Keshara purchased maize flour to be used to prepare porridge for the family for breakfast at their home and its known that they need between 200g to 300g mass of maize flour to be mixed with soya and milk but on reaching home, the mother doubted if the flour was of the required mass, and needed to confirm but didn't have the sufficient knowledge to do so, so she approached you for help.

TASK:

You are provided with a metre rule and an object whose quantity is equal to that of the maize flour purchased by Keshara, use it to advise her mother.

ITEM SIXTEEN (16)

An investor owns a factory that manufactures gum boots. He buys rubber at sh.500 per kilogram. A girl collects 1000 rubber bungs whose masses are in the range 50 g – 100g inclusive as required by the investor. The girl would like to know how much she can earn from the sale of one rubber bung but has no weighing scale to measure its mass.

TASK:

As a student of physics, carry out a scientific investigation to help the girl determine how much she can get from the sale of one rubber bung.

ITEM SEVENTEEN (17)

A student had a problem of seeing near objects clearly. When he went to an optician he was told to buy spectacles to aid his vision. Unfortunately, the glasses fell down and the lens in the glass broke. he decided to visit the nearby eye clinic but did not know the focal length of the lens in his broken spectacles. The broken spectacles had a lens whose focal length was in the range 9.0cm – 11.0cm. You have been provided with a lens similar to the one that got broken in the student's spectacles.

TASK:

As a learner of physics, carry out a scientific investigation to determine whether the lens bought can be used to replace the broken one.

ITEM EIGHTEEN (18)

A wholesaler dealing in dry cells checks their quality by considering their internal resistance to be 0.5Ω after manufacturing for a good quality otherwise they are of poor quality. The wholesaler receives the consignment but the person in charge of checking their internal resistance is away and the wholesaler is challenged whether the cells are of good quality.

TASK:

As a learner of physics, help the wholesaler to know whether the dry cells delivered are of good quality.

HINT: {Include the following in your report: Aim, variables (Independent, dependent and controlled), hypothesis, Procedure, Errors anticipated, Precautions to mitigate the errors, Present data accurately (at least 5 sets), Analyse data (using graph) and advice/recommend }.

ITEM NINETEEN (19)

In one of the schools, during project work, the S.4learners opted to make a reflecting telescope. All the other materials were available except the concave mirror of radius of curvature in the range of 18.0cm and 22.0cm inclusive. One of the learners bought a concave mirror but there was no label on it.

TASK:

you are provided with one concave mirror from the laboratory with similar properties like the one the learner bought, sensitize the S.4 learners whether to use it or not.

ITEM TWENTY (20)

A certain organization has donated juice making machine to one of the farmers dealing in fruit business. The farmer plans to stock juice in bottles with tight covers made of rubber. The recommended type of rubber should have density that lies in the range 0.9gcm-3to 1.1gcm-3. The farmer was advised to design the cover using the rubber bung, but not sure whether specification of the rubber material will meet the required standard.

TASK:

Use the rubber bungs in your physics laboratory to confirm whether it is the type the farmer intends to acquire to be used as bottle covers.

ITEM TWENTYONE (21)

A manufacturer is designing an electric heater whose efficiency depends on the resistance of the heating element from a nichrome wire. The heater is expected to produce an output of 1000w when connected to a 240v ac supply.

TASK:

As a learner of physics, find out the possibility of using the nichrome wire which is in the laboratory.

ITEM TWENTYTWO (22)

The projector of the school has been brought to a workshop where you volunteer as a technician because its converging lens whose power is between 100D and 300D has got broken.

TASK:

Using scientific investigation, help the ICT teacher as a learner of physics to find out whether the lens available in the Physics laboratory can be used to fix the projector or not.

ITEM TWENTYTHREE (23)

Your friend has just bought a quarter kilogram of sugar from a shop keeper who she thinks has given her less sugar yet has no weighing scale but there is a standard mass of 100g.

TASK:

Using your knowledge of Physics and assuming the principle of moments applies, carry out scientific investigation to find out the right quantity of the sugar.

ITEM TWENTYFOUR (24)

In one of secondary schools in the central region, the laboratory technician in charge of the biology laboratory went to the village for burial, unfortunately this happened when senior one biology class were to use microscopes to study in depth features of a leaf. An effort to get the copy of keys from head of department hit a dead end since he was equally out station. This prompted the biology teacher to consult other departments where he was advised to improvise by using convex lens of magnifying power of 10 D. Physics laboratory has plenty of such lenses but are mixed with other lenses of different specification and are all not labelled.

TASK:

As a physics student, carry out a scientific investigation to sort the lenses such that lesson does not fail.

ITEM TWENTYFIVE (25)

A battery manufacturing company has received complaints of its products overheating, and having low life span. This made the company director to have a consultation with its team of experts' in-charge of quality assurance; they predicted that this could be as result of internal resistance of the batteries exceeding 1.8 Ω and proved it right. In a bid to improve on the quality and life span of the batteries, the company's executive director decided that all the newly manufactured batteries should be tested before putting them into the market. Unfortunately, after the commencing the process of testing, the machine broke down. This forced them to seek for alternative approach which they are not familiar with, they seek for your expertise.

TASK:

Carryout scientific investigation to determine whether their newly manufactured batteries meet the minimum standards.

ITEM TWENTYSIX (26)

One of the items that boy children like playing with is a toy gun. To please his child, a father bought a toy gun as birthday gift. In the evening, while the child was playing with his friends, the toy gun stopped working and the child started crying. When the father saw his child crying, he picked up the toy gun and took it to a mechanic for repair. On opening the toy gun, the mechanic found out that the spring in the toy gun, of force constant 20Nm-1 was damaged. The father asked the mechanic to replace the damaged spring with another suitable spring, but the only available spring, had no specifications.

TASK

As a learner of physics carry out a scientific investigation to determine whether the available spring is suitable.

ITEM TWENTSEVEN (27)

A mobile money operator working along one of the busy streets in Kampala faced a challenge one day when the Umbrella she was seated under got blown away by a violent wind. She picked the broken umbrella a few metres away from her work station but discovered that its springs were completely destroyed. When she took the umbrella for repair, the mechanic requested her to buy new springs of force constant 25 Nm^{-1}, which she did. There was, however, no label on the springs she bought indicating the value of the spring constant.

TASK:

You are provided with a spring X having the same properties as those of the spring bought by the mobile money operator. Carry out a scientific investigation to determine the force constant of the spring.

ITEM TWENTYEIGHT (28)

In most of Uganda's homes, charcoal is the main source of heat energy for cooking. Because there is high demand for charcoal, many trees have been felled, leading to deforestation and an increase in global warming. To mitigate the ever-increasing effects of global warming, experts running a Non – Governmental Organization (NGO) advised people to turn to alternative sources of energy like solar, gas, electricity etc. The NGO donated concave mirrors of focal lengths in the range 15.0 cm – 20.0 cm to be used to harness solar energy from the sun. The challenge is finding the solar heating points of the mirrors.

TASK:

As a student of physics, carry out a scientific investigation to locate the solar heating point of a concave mirror donated to your home.

ITEM TWENTYNINE (29)

A box of new torch bulbs was delivered at a school laboratory for use in practical physics investigations. The new bulbs looked identical to an old stock of torch bulbs that had been supplied the previous term. When each of the new bulbs was connected to a new dry cell, the light emitted was not as bright as that obtained from each of the bulbs that were delivered earlier. This led to the suspicion that the resistance of the filament in the new stock of torch bulbs was higher than the recommended value of 1.0 Ω.

TASK:

As a student of physics working closely with the laboratory technician at your school, carry out a scientific investigation to determine the resistance of one of the new bulbs supplied.

ITEM THIRTY (30)

A certain number of metre rules of masses in the range of 78g to 140g are to be organised and transported to the laboratory where an experiment is to be performed. In the store where the metre rules are kept the beam balance is not available, but a standard mass are available.

TASK:

Advise the store keeper on how to determine the mas of each metre rule.

ITEM THIRTYONE (31)

A rich man constructed a house and on completion, the glasses that were fixed in the windows consistently kept breaking. As a result, he sought advise from an expert who told him that the require range for refractive index of the glass is 1.4 – 1.6. The expert showed him a sample of the glass that he would have to buy.

TASK:

As a learner of physics, help the rich man to choose the correct glass to use.

ITEM THIRTTWO (32)

One afternoon, strong winds blew off the trader's umbrella. The mechanic discovered that the spring of force constant in the range of 40 Nm^{-1} to 100 Nm^{-1}, in the umbrella was destroyed and needed replacement. The trader bought a new spring without specification from a nearby hardware, and approached you for help.

TASK:

You are provided with a spring from the laboratory, with similar properties as the one bought by the trader and whose force constant is half of that in the umbrella, advise the trader whether it can be used to replace the one in the umbrella.

HINT: Include the following in your report; Aim, variables (independent, dependent and controlled), hypothesis, procedure, errors anticipated, precautions to mitigate the errors, present data accurately (at least 5 sets), analyse data (using graph) and advice/recommend.

ITEM THIRTYTHREE (33)

A builder fixing windows for a community member proposed that the best glass for the window shutter should be of critical angle between 38° and 42° inclusive. The community member bought a large sheet of glass from a nearby shop that sells glasses but was not sure of its critical angle and contacted you for assistance.

TASK:

You are provided with a glass block from the laboratory, with similar properties as the large sheet of glass bought by the community member, advise accordingly on its suitability for window shutter.

HINT: Include the following in your report; Aim, variables (independent, dependent and controlled), hypothesis, procedure, errors anticipated, precautions to mitigate the errors, present data accurately (at least 5 sets), analyse data (using graph) and advice/recommend.

ITEM THIRTYFOUR (34)

A teacher of physics sent a class representative to a laboratory attendant to collect convex lenses whose focal lengths range from 8.0cm t0 12.0cm inclusive, for use during a class demonstration.

The attendant found it difficult to select the lenses of the required specifications since there were many unlabelled boxes in the laboratory, containing lenses.

The attendant randomly picked one of the lenses that was in the box received by the teacher.

TASK:

Design and carry out an investigation to determine whether the lens is suitable for use in the class demonstration.

HINT:

In your report, include, the aim, variables (independent, dependent and controlled), hypothesis, procedures, possible sources of error, precautions to the stated errors, data presentation (at least 5 data sets), data analysis(using graph) and advise/ recommendations.

ITEM THIRTYFIVE (35)

A baby playing with a toy car which only operates on a cell of e.m.f ranging from 1.2V to 1.7V, started crying when it stopped to move. On a closer observation, the parents found out that the e.m.f of the dry cell in the toy had dropped, and decided to buy a new one. Since the writing on the cell could not be seen well, the parent does not know whether the new battery bought has the right specification.

You are provided with the new cell the parent bought.

TASK:

Design and carry out an investigation to determine if the dry cell bought can be used in the toy car.

HINT:

In your report, include, the aim, variables (independent, dependent and controlled), hypothesis, procedures, possible sources of error, precautions to the stated errors, data presentation (at least 5 data sets), data analysis(using graph) and advise/ recommendations.

ITEM THIRTY SIX (36)

Your brother wants to equip the Physics laboratory of his school with concave mirrors. He wishes to buy concave mirrors of focal lengths between 9.0 cm and 11.0 cm inclusive. Your brother has received samples of concave mirrors from a supplier who claims that they are of the required range of focal lengths. However, your brother wants to confirm the supplier's claims before buying the concave mirrors but lacks the knowledge.
You have been provided with the sample of the concave mirror that was delivered to your brother.

TASK

Determine whether or not your brother should buy the concave mirrors.

ITEM THIRTY SEVEN (37)

During school project work, you came up with an idea of making your own dry cell, whose quality is comparable to that of the dry cells on the open market. Through research, you found out that the quality of a dry cell is affected by its internal resistance.
The dry cells on the market are of e.m.f 1.5 V. However, their internal resistances are not indicated. You wish to determine the internal resistance of one of these cells, so that you use it as a standard when making yours.
Your teacher has provided you with a sample of a new dry cell on the market.

TASK

Determine the internal resistance you will base on when making your cell.

ITEM THIRTY EIGHT (38)

The Physics Department of your school uses glass blocks for practical work. All the glass blocks in the laboratory have widths between 6.0 cm to 7.0 cm. During a laboratory session, one of the glass blocks was accidently broken and a replacement was purchased. However, the dimensions of the replacement glass block were not indicated on its package.
You are provided with the replacement glass block.

TASK
Design and carry out an investigation to determine whether the width of the glass block lies within the acceptable range.
Hint

In your report, include; the aim, variables (independent, dependent and controlled), hypothesis, procedure, possible sources of errors with corresponding mitigations, at least 5 data sets, graphical analysis of data, and conclusion/advice.

ITEM THIRTY NINE (39)
A snack vendor uses a heat-sealing machine to package products for school canteens. Recently, customers complained about a drop in snack quality due to poorly sealed, non-airtight packaging. An inspection revealed that the machine's heating element is broken.
To fix the machine, the vendor needs to install a new heating element using a resistance wire. For the machine to operate safely and effectively, this wire must have a resistance per unit length between 8.0 Ω to 12.0 Ω. The vendor possesses an unidentified roll of wire but lacks the technical knowledge to check if it meets these specifications.
You are provided with a sample of wire from the vendor's roll, along with standard laboratory equipment.
Task
Design and execute an experimental investigation to test the wire sample.
Hint
In your report, include; the aim, variables (independent, dependent and controlled), hypothesis, procedure, possible sources of errors with corresponding mitigations, at least 5 data sets, graphical analysis of data, and conclusion/advice.

ITEM FORTY (40)
A school wants to equip the physic laboratory with concave mirrors of focal lengths between 9.0cm to 11.0cm inclusive to be used in mock examinations. The director of studies received samples of concave mirrors from the supplier who claims that they are of the required range of focal length. However, the director of studies wants to confirm the supplier's claim before buying the concave mirrors but lacks the knowledge. You have been provided with the samples of the concave mirrors that were delivered to the director of studies.
TASK:
Design and carry out scientific investigations to determine the focal length of the concave mirror and whether the school should buy them or not.

ITEM FORTY ONE (41)
A certain home owner ordered for dry cells each of ***emf*** ranging between 1.3V to 1.7V and internal resistance ranging from 0.2Ω to 1.2Ω to be used in lighting the bulb in his flash light. When the cell was delivered to him, he observed that there were no specifications marked on it and was not convinced that it was the right cell to be used in his flash light. The home owner would like to establish whether the dry cell received was of required specifications before payment could be done. You have been provided by the dry cell received by the home owner.
TASK:
Design and carry out a scientific investigation to determine the emf and internal resistance of the dry cell and advise the home owner whether to buy or not.
Hint
In your report, include; the aim, variables (independent, dependent and controlled), hypothesis, procedure, possible sources of errors with corresponding mitigations, at least 5 data sets, graphical analysis of data, and conclusion/advice.

ITEM FORTY TWO (42)
The gardener uses hedge scissors whose force constant is between $20Nm^{-1}$ and $50Nm^{-1}$ to cut the hedge around the school compound, observed that the scissors were no longer providing the efficient force to cut the hedge easily.
He took it to the Head teacher so that a new one could be bought. The head teacher was

advised by the technician to change the spring in the scissors.
The technician recommended for the replacement of the spring with another one of the same force constant range. The Head teacher found the available springs in the shop were not marked with the required specifications.
You are provided with the same spring used in the scissors.

TASK;

Design and carry out scientific investigation to determine the force constant of the spring and whether the Head teacher should go on and buy it.

ITEM FORTY THREE (43)

A senior three learner in your school has been complaining that she cannot see clearly on the black board, whenever she sits in front in class.
She was advised by the school nurse to buy spectacles with convex lenses of power ranging from 8D to 12D. After buying the spectacles from the nearby shop, she was not sure if they could serve the intended purpose and approached you for help.

Task:

As a physics learner, you are provided with a lens from the laboratory which has the same properties as those of the spectacles bought, Design and carry out a scientific investigation to determine the power of the spectacles and whether they should be used.

HINT:

In your report, include; the item, variables (Independent, dependent, and controlled) hypothesis, procedure, possible sources of errors, precautions to the stated errors, data presentation (at least 5 data sets), data analysis (using graph) and advise/recommendation.